Sebastian De Leon

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Summary

Game Developer and Computer Engineering student specializing in gameplay programming, 3D math, physics simulation, and real-time interactive systems. Engine- and language-agnostic engineer experienced in architecting decoupled, data-driven systems across Unity, Godot, WebGL/Three.js, and custom C++/C# frameworks, with a track record of winning hackathons and shipping high-performance interactive titles.

Technical Skills

- **Core Architecture**: Component-Based Architecture (ECS principles), State Machines (FSM), Object Pooling, Event-Driven Systems, Game Loops, Data-Driven Design, Input Buffering
- **Math & Physics**: Linear Algebra, Vector Math, Quaternions & Rotations, Rigid-Body Dynamics, Kinematics & Inverse Kinematics (IK), Collision Resolution, Raycasting, Buoyancy Modeling
- **Languages & Shaders**: C++, C#, GDScript, C, TypeScript, JavaScript, Python, GLSL, SQL | English (Fluent), Spanish (Native)
- **Engines & Frameworks**: Unity, Godot, WebGL / Three.js, Custom Systems, React, .NET Core
- **Tools & Pipelines**: Git, Blender, Blockbench, Unity Profiler, Frame Debugger, Visual Studio, VS Code, Linux

Featured Projects

NASA HERC Rover Simulator — Unity, C#, WebGL
Physics-Driven 3D Planetary Simulation — Live WebGL Simulation
- Engineered a 3D vehicle simulation modeling multi-body suspension physics, wheel colliders, terrain friction, and multi-axis actuator kinematics across simulated Martian and Lunar topography.
- Implemented vehicle finite state machines, mechanical task interaction triggers, and a real-time HUD telemetry dashboard; profiled draw calls and physics tick rates to maintain a locked 60 FPS in browser viewports.

Ying Yang Cats — Godot, GDScript, WebGL
Dual-Character Synchronized Puzzle Platformer — WMU Hackathon Winner (2025)
- Architected a dual-character control pipeline managing simultaneous inverted gravity vectors, input buffering, and tight platformer physics across twin synchronized viewports.
- Implemented custom collision resolution, movement state machines, and cooperative puzzle triggers; awarded **1st Place overall** at the WMU Computer Club Annual Hackathon / Game Jam.

Marine Guardians (Guardianes Marinos) — Unity, C#, WebGL
3D Ocean Exploration & Simulation — Shipped WebGL Title
- Programmed 6-DOF underwater locomotion physics with non-linear buoyancy vectors, hydrodynamic drag, and dynamic oxygen consumption mechanics.
- Designed a decoupled, data-driven wildlife encyclopedia using ScriptableObjects, enabling rapid iteration of creature attributes, sound triggers, camera damping, and atmospheric shaders.

Printune — Three.js, TypeScript, Python
Procedural 3D Mesh Generator & WebGL Engine — Live Web Application
- Created a procedural 3D mesh engine generating manifold, watertight geometries from 2D vector data with customizable extrusions, bevels, and materials in real time.
- Implemented client-side 3D scene graphs, orbit camera controls, lighting shaders, and direct browser-based binary STL geometry serialization.

Experience

Interactive Systems Developer / Telemetry Assistant — Aug 2024 – Apr 2025
Apolo 27 (NASA HERC College Team) — Santo Domingo, Dom. Rep.
- Directed the development of proprietary interactive games and simulation tools for national STEM outreach across the Dominican Republic, demonstrating aerospace mechanics and robotics concepts.
- Engineered the rover embedded control system around an ESP32-S3, implementing finite state machines (FSM) to map real-time PWM controller inputs to kinematic motion profiles and multi-axis actuators.
- Contributed to winning **1st Place Overall** at the NASA Human Exploration Rover Challenge (RC Division) through reliable obstacle traversal, telemetry, and control systems.

Junior Software Engineer — Jul 2024 – Jul 2026
Reispsolutions S.R.L. — Santo Domingo, Dom. Rep.
- Engineered production services using Clean Architecture, DDD, and CQRS in .NET 9, enforcing strict separation of concerns, data-driven domain models, and zero unnecessary runtime allocations.
- Developed interactive Android kiosk client applications with fluid state-driven animations, gesture inputs, and decoupled event architectures; profiled runtime performance to ensure responsive frame execution.

Co-Founder & Technical Lead — Nov 2022 – Jul 2025
DreamBytes Game Development Club (INTEC) — Santo Domingo, Dom. Rep.
- Co-founded the university's premier student game development club; organized collegiate game jams and hackathons across multiple engines (Unity, Godot, custom engines).
- Delivered technical workshops on gameplay programming, 3D vector math, component architectures, and game loop optimization for aspiring student developers.

Education

Western Michigan University — Sep 2025 – Dec 2027
B.S. in Computer Engineering, GPA: 3.89 | CEAS Dean's List (all semesters) — Kalamazoo, MI

Instituto Tecnológico de Santo Domingo — Nov 2022 – Jul 2025
B.E. in Computer Science (transferred), GPA: 3.89 | Co-Founder of DreamBytes Game Dev Club — Santo Domingo, Dom. Rep.